

Pablo Busch

Ph.D. Energy Systems, M.Sc. Statistics & Public Policy, Industrial & Environmental Engineer
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ACADEMIC APPOINTMENTS

Pontificia Universidad Católica de Chile Assistant Professor - Civil Engineering & Institute for Sustainable Development	Santiago, Chile <i>Aug. 2026 – Present</i>
Stanford University Postdoctoral Scholar – Stanford Doerr School of Sustainability & Stanford STEER • Sustainable Mining Supply Chains • Global Material Footprint	Stanford, CA <i>Sep. 2025 – Jul. 2026</i>

EDUCATION

University of California, Davis Ph.D. Energy Systems • Quantitative modelling of Demand and Supply for Energy Transition Minerals M.Sc. Statistics - Data Science • Outstanding Master's Student Award. GPA: 4.0 M.Sc. Environmental Policy and Management • Environmental & Resource Economics • Spatial Information Science	Davis, CA <i>Jan. 2023 – Jun. 2025</i> <i>Sep. 2021 – Dec. 2022</i> <i>Sep. 2020 – Jun. 2022</i>
Pontificia Universidad Católica de Chile Civil Engineer of Industry, Diploma in Environmental Engineering • Title conferred with maximum distinction • Award for Best Student of the Hydraulic and Environmental Engineering Department • Honor award, Highest GPA of the class (2014 and 2015)	Santiago, Chile <i>Mar. 2011 – Sep. 2016</i>

PROFESSIONAL EXPERIENCE

Researcher University of California, Davis • Institute of Transportation Studies & Energy and Efficiency Institute * Developed a global model of critical mineral demand and supply to evaluate supply risks for lithium-ion battery decarbonization technologies * Developed a global trade supply chain model for electric vehicle production and policy scenarios * Analyzed lifecycle emissions assessment of hydrogen pathways to zero-carbon emission vehicles in California * Evaluated decarbonization strategies and policies for the cement & concrete sector	Apr. 2021 – Aug. 2025 <i>Davis, CA</i>
Policy Analyst Chile Lagos Limpios • Summarized policy lessons and impact metrics from Lake Tahoe conservation experience to protect North-Patagonian lakes	Sep. 2021 – Jun. 2022 <i>Davis, CA - Valdivia, Chile</i>
Senior Project Engineer GreenLab, Dictuc S.A. • Air Quality * Conducted an ecological study to analyze the relationship between air pollution (PM_{2.5}), derived mainly by wood combustion, and COVID-19 mortality rates in Chile * Designed the methodological guide for estimating atmospheric emissions from point sources in Chile * Executed a technical and economic analysis to implement an environmental policy instrument to improve air quality in the inner zone of the Valparaíso region • Life Cycle Assessment * Designed and directed the development of a web tool of life cycle assessment in the packaging sector, using local data from the industry • Health & Risk Assessment * Developed an automated model to calculate the national base incidence rates of mortality and morbidity for air pollution in Chile * Summarized the main risk perception challenges of hydrogen combustion technologies in the mining sector	Aug. 2018 – Nov. 2020 <i>Santiago, Chile</i>

Project Engineer

GreenLab, Dictuc S.A.

Sep. 2016 – Jul. 2018

Santiago, Chile

- **Climate Change & Air Quality**

- * Calculated the first national emissions inventory of short lived climate pollutants in Chile
- * Designed a handbook for the elaboration of emissions inventory in Chile
- * Improved the methodology for the estimation of health benefits derived from atmospheric pollution reductions

- **Environmental Economics**

- * Executed an econometric analysis for a contingent valuation study to estimate environmental benefits associated with adequate disposal of car waste, such as tires, motor oils and batteries
- * Conducted a specialized technical assistance in matters of information analysis of environmental inspection activities

- **Life Cycle Assessment**

- * Analyzed the social and economic impacts of a return and refund system on returnable containers for soft drinks

Research Internship

Technische Universität Berlin

Jan. 2016

Berlin, Germany

- Analyzed methodologies for life cycle assessment

SKILLS

Languages: Spanish Native, English Proficient, German Intermediate

Programming: R, julia, Python, SQL, JavaScript (D3), Git, L^AT_EX, VBA

Software: Visual Studio Code, Positron, R Studio, ArcGIS, QGIS, Tableau, Stata, Analytica

SCIENTIFIC ARTICLES

Web of Science Publications (Google Scholar: h-index = 9, citations = 497)

1. **Busch, P.**, Chen, Y., Kendall, A. (2026). Life cycle performance and carbon handprint of lithium-ion batteries in electric vehicles. *Journal of Industrial Ecology*.
2. Santero, N., Nelson, L., Chen, Y., Meredith, M., **Busch, P.**, Kendall, A. (2026). Electrifying light vehicles in the United States shows emission reduction potential for all vehicle types and powertrains. *Communications Sustainability*.
3. Duex, S., **Busch, P.**, Kendall, A. (2026). The electric vehicle transition: effects on copper supply dynamics in a net-zero future. *Resources, Conservation and Recycling*, 228, 108798.
4. **Busch, P.**, Chen, Y., Ogbonna, P., Kendall, A. (2025). Effects of Demand and Recycling on the When and Where of Lithium Extraction. *Nature Sustainability*, 1-11.
5. **Busch, P.**, Pares, F., Tal, G., Kendall, A. (2025). Who Will Have Enough? Battery Mineral Demand and Sufficiency in Vehicle Producing Countries. *Environmental Science & Technology*, 59(45), 24392-24403
6. Chandra, M., **Busch, P.**, Pares, F., Tal, G. (2025). Paths of progress: Forecasting global electric vehicle demand amidst demographic and economic growth. *Transportation Research Part D: Transport and Environment*, 147, 104928
7. **Busch, P.**, Kim, A., Miller, S.A., Murphy, C.W. (2025). Policy mechanisms to decarbonize cement production: through the lens of California. *Environmental Research: Infrastructure and Sustainability*, 5(2), 025003.
8. Pares, F., **Busch, P.**, Chandra, M., Tal, G. (2024). Shifting Manufacturing: Electric Vehicle Supply Strategy using the Model for International EV Trade. *Journal of Cleaner Production*, 144357.
9. **Busch, P.**, Rocha, P., Lee, K., Cifuentes, L., Tai, X. (2024). Short-term exposure to fine particulate pollution and elderly mortality in Chile. *Communications Earth Environment*, 5(1), 469.
10. **Busch, P.**, Pares, F., Chandra, M., Kendall, A., Tal, G. (2024). Future of Global Electric Vehicle Supply Chain: Exploring the Impact of Global Trade on Electric Vehicle Production and Battery Requirements. *Transportation Research Record*, 2678(11), 1468-1482.
11. Brandao, M., **Busch, P.**, Kendall, A. (2024). Life cycle assessment, quo vadis? Supporting or deterring greenwashing? A survey of practitioners. *Environmental Science: Advances*. 3(2), 266-273
12. **Busch, P.**, Kendall, A., Lipman, T. (2023). A systematic review of life cycle greenhouse gas intensity values for hydrogen production pathways. *Renewable and Sustainable Energy Reviews*, 184, 113588.
13. **Busch, P.**, Cifuentes, L.A., Cabrera, C. (2023). Chronic exposure to fine particles (PM_{2.5}) and mortality: Evidence from Chile. *Environmental Epidemiology*. e253.
14. **Busch, P.**, Kendall, A., Murphy, C.W., Miller, S.A. (2022). Literature review on policies to mitigate GHG emissions for cement and concrete. *Resources, Conservation and Recycling*, 182, 106278.

Other Publications

- Busch, P.**, Pares, F., Kendall, A., Sulwe, C. (2026). Batteries Without Borders: Trade, Circularity, and the Future of Critical Mineral Security. *Majority Mobility - Issue 2*.
- Pares, F., **Busch, P.** (2024). Renewable Energy and Energy Storage Value Chains in Latin America and the Caribbean. *Inter - American Development Bank*.
- Lipman, T., **Busch, P.**, Collins, S., Horvath, A., Kendall, A., Coffee, D., Kong, D. (2024). Moving Beyond the Colors: The Full Life-Cycle Emissions of Hydrogen Production Pathways for California.
- Cheng, K., **Busch, P.**, Ziotopoulou, K. (2024). Explainable Machine Learning Interpretations on New Zealand Random Forest Liquefaction Manifestation Predictions.
- Tal, G., Pares, F., **Busch, P.**, Chandra, M. (2023). Implications of Global Electric Vehicle Adoption Targets for the Light Duty Auto Industry in Mexico.
- Lipman, T., Horvath, A., Collins, S., Kendall, A., Fulton, L., **Busch, P.** (2022). Hydrogen Can Have a Much Lower Carbon Intensity than Fossil Fuels But This Largely Depends on How It Is Produced and Distributed.
- Busch, P.**, Kendall, A., Murphy, C. W., Miller, S. (2022). Industrial Decarbonization: Policy Pathways for the Cement Concrete Sector.

Conferences or Seminars Talks

- International Symposium on Sustainable Systems and Technology 2026 *Preventing scarce water use from battery minerals extraction*. June 2026.
- ETH Zurich Academy on Sustainability and Technology 2026 *Global material consumption persists despite optimal efficiency and circularity*. May 2026.
- 12th International Conference on Industrial Ecology. *Poster on Effects of Demand and Recycling on the When and Where of Lithium Extraction*. July 2025.
- Global Forum for Sustainable Batteries, UC Berkeley Law. *More lithium recycling, less mining*. July 2025.
- WWU Energy Speaker Series. *Preventing unnecessary mine openings in the transition to clean energy*. April 2025
- Transportation Research Board 103rd Annual Meeting. *Future of Global Electric Vehicle Supply Chain*. January 2024.
- 19th Asilomar biennial conference on Transportation & Climate Policy. *Future of Global Electric Vehicle Supply Chain*. July 2023.
- Centro Innovacion UC. *Good practices and circular economy strategies for sustainable packaging production*. 2018
- II Chilean Conference on Life Cycle Assessment. *Web tool to conduct life cycle assessment in the packaging sector*. 2018

ACADEMIC EXPERIENCE

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| Scholar <i>Resources for the Future</i> | Sep. 2024 – Jun. 2025 |
| <ul style="list-style-type: none">Research scholar at the Critical Minerals Research Lab, an interdisciplinary collaboration to ensure a reliable and equitable supply chain for critical minerals | |
| Committee Member <i>ENAMI, Chile</i> | May – July 2024 |
| <ul style="list-style-type: none">Participation as expert on the Chilean Committee for new Technologies for Lithium Extraction | |
| Consultant <i>Inter American Development Bank</i> | March – June 2024 |
| <ul style="list-style-type: none">Report on emerging energy supply chain opportunities for Latin America and the Caribbean | |
| Conference Presentation <i>TRB, Washington DC</i> | Jan. 2024 |
| <ul style="list-style-type: none">Future of Global Electric Vehicle Supply Chain: Exploring the Impact of Global Trade on Electric Vehicle Production and Battery Requirements | |

Awards

- Dissertation Grant:** National Center for Sustainable Transportation University Transportation Center. Award: \$20,000 USD to complete dissertation titled "A quantitative model of critical mineral supply and demand for a clean energy system".
- Most Downloaded Paper:** Article "Literature review on policies to mitigate GHG emissions for cement and concrete" was the most downloaded paper from 2022-2024 in the journal *Resources, Conservation Recycling (RCR)*.